MAJOR PROJECT PRESENTATION

Identification of Skin Abnormalities via Deep Learning: An Optimized ResNet50 Approach

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Problem Statement

- Skin diseases are common but often misdiagnosed or detected late as we don’t take them seriously in the beginning until it looks serious.
- Limited access to expert dermatologists in many areas and also the lack of medical facilities in remote areas.
- Manual diagnosis is time-consuming and subjective.
- Generally new deep learning models are computationally expensive and require huge amount of storage.
- Need for an accurate, fast, and optimized detection system for better result
- Early detection of skin diseases is challenging and error-prone.
- Traditional methods depend heavily on expert availability

Objective

The main objective of our project is to classify 22 different skin diseases using an optimized ResNet-50 model, improve accuracy compared to traditional CNN models, and provide an automated system that reduces dependency on manual diagnosis.

- Classify skin images into **22 disease categories**
- Enhance model performance through **CNN optimization techniques,** using transfer learning, data augmentation etc.
- Reduce manual effort by providing **automated and reliable diagnosis:**

  Input Image → Preprocessing → Model (ResNet-50) → Prediction → Result

Why Deep Learning ?

- Automatically extracts **complex features** from images, as it has multiple segments like:

  Early layers → edges, colors

  Middle layers → textures, patterns

  Deep layers → disease-specific features

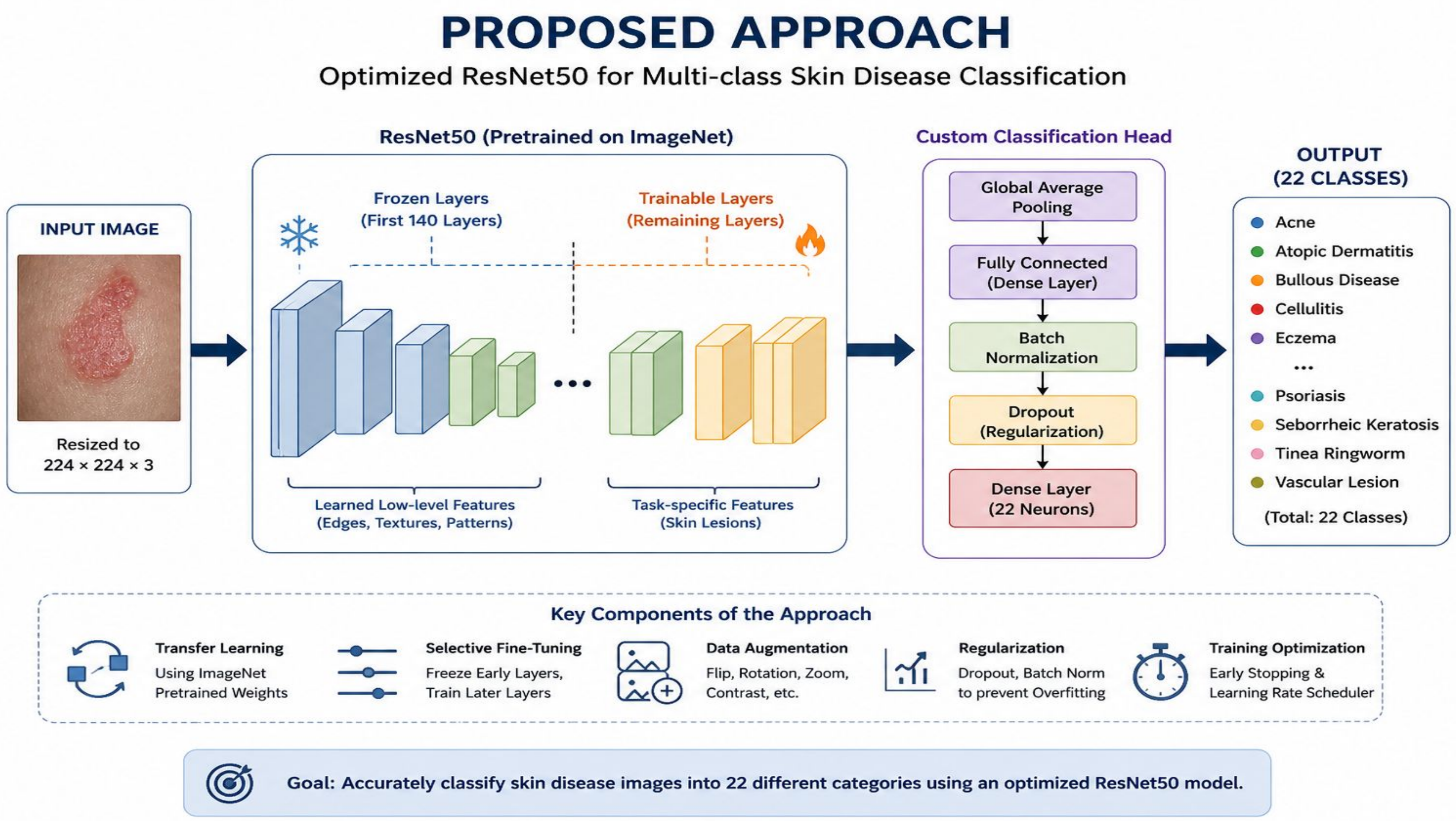
- Outperforms traditional machine learning in **image classification tasks**
- Handles **large and diverse datasets** effectively because it automatically learns to extract complex features at different level.
- Provides **high accuracy and robustness** because of abstraction, data augmentation, fine tuning.

Proposed Approach

Our proposed approach is based on ResNet50, which is a deep convolutional neural network **pretrained on ImageNet** (Large scale database designed for computer vision research). Instead of training from scratch, we **used transfer learning** to reuse the learned features.

We froze the some initial layers to preserve low-level features like edges and textures, and fine-tuned the remaining layers to adapt to skin disease classification.

On top of the base model, we added a custom classification head with global average pooling and dense layers to classify images into 22 disease categories


PROPOSED APPROACH
Optimized ResNet50 for Multi-class Skin Disease Classification
INPUT IMAGE
Resized to
224 × 224 × 3
ResNet50 (Pretrained on ImageNet)
Frozen Layers
(First 140 Layers)
Trainable Layers
(Remaining Layers)
Learned Low-level Features
(Edges, Textures, Patterns)
Task-specific Features
(Skin Lesions)
Custom Classification Head
Global Average Pooling
Fully Connected (Dense Layer)
Batch Normalization
Dropout (Regularization)
Dense Layer (22 Neurons)
OUTPUT
(22 CLASSES)
Acne
Atopic Dermatitis
Bullous Disease
Cellulitis
Eczema
...
Psoriasis
Seborrheic Keratosis
Tinea Ringworm
Vascular Lesion
(Total: 22 Classes)
Key Components of the Approach
Transfer Learning
Using ImageNet Pretrained Weights
Selective Fine-Tuning
Freeze Early Layers, Train Later Layers
Data Augmentation
Flip, Rotation, Zoom, Contrast, etc.
Regularization
Dropout, Batch Norm to prevent Overfitting
Training Optimization
Early Stopping & Learning Rate Scheduler
Goal: Accurately classify skin disease images into 22 different categories using an optimized ResNet50 model.

Methodology

First, we collect the dataset and **preprocess the images**. Then we apply **data augmentation** to improve generalization. The processed data is passed to the ResNet-50 model for training. After training, we test and evaluate the model using performance metrics. Finally, the system predicts the skin disease from a new input image.

Dataset → 22 skin disease images

Preprocessing → Resize, Normalize

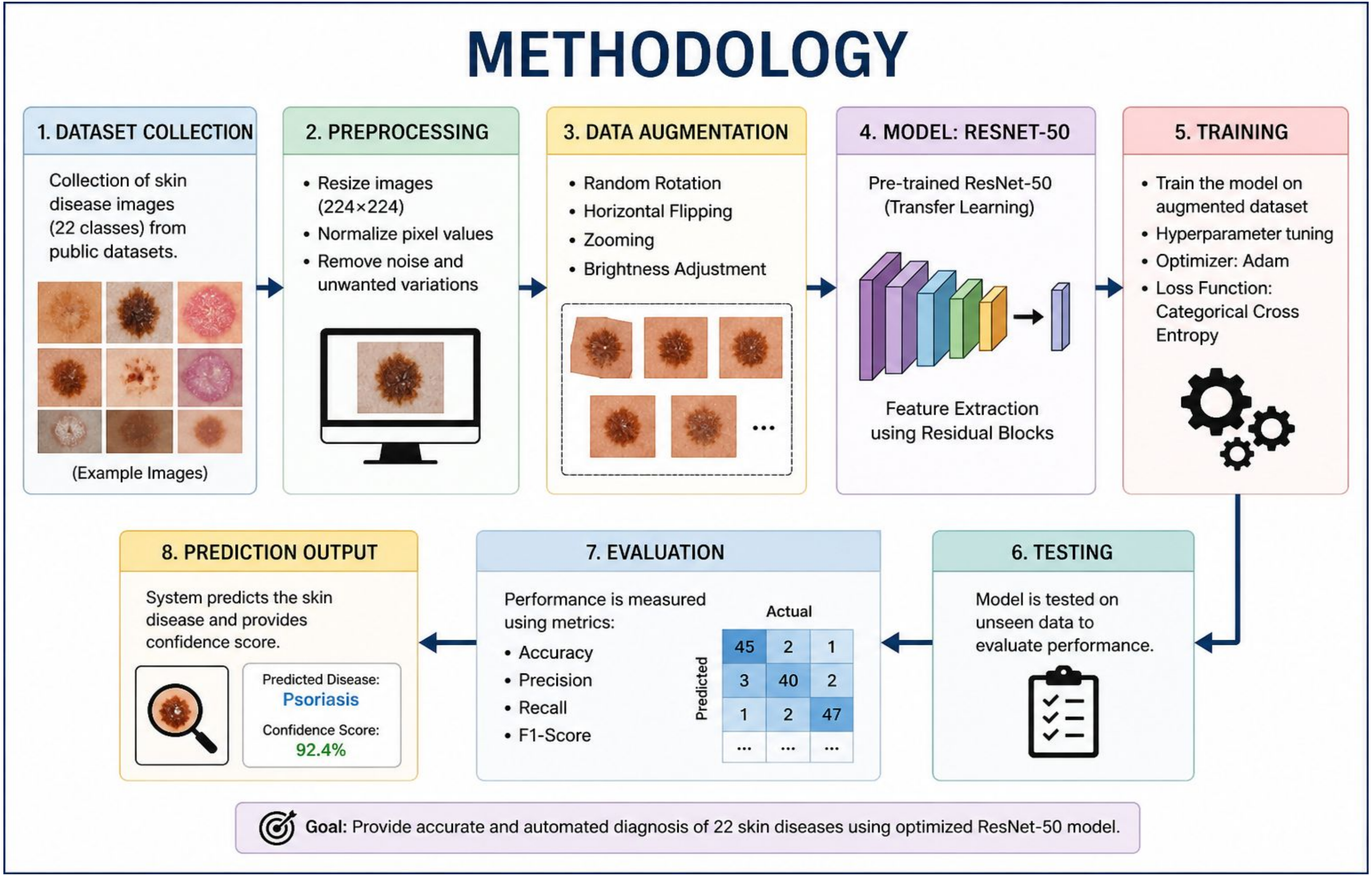
Augmentation → Flip, Rotate, Zoom

Model → ResNet-50 (Transfer Learning)

Training → Optimize model

Evaluation → Accuracy, Precision, Recall

Output → Predicted disease + confidence

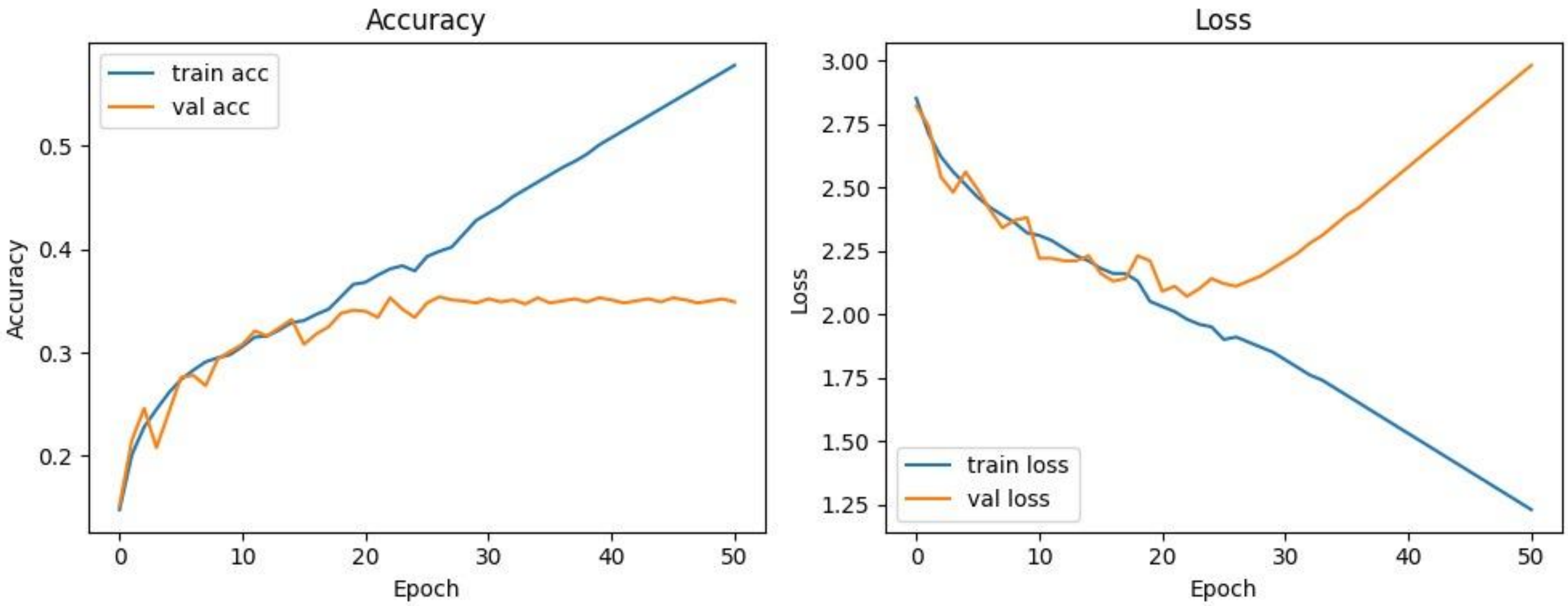

METHODOLOGY
1. DATASET COLLECTION
Collection of skin disease images (22 classes) from public datasets.
(Example Images)
2. PREPROCESSING
• Resize images (224×224)
• Normalize pixel values
• Remove noise and unwanted variations
3. DATA AUGMENTATION
• Random Rotation
• Horizontal Flipping
• Zooming
• Brightness Adjustment
4. MODEL: RESNET-50
Pre-trained ResNet-50 (Transfer Learning)
Feature Extraction using Residual Blocks
5. TRAINING
• Train the model on augmented dataset
• Hyperparameter tuning
• Optimizer: Adam
• Loss Function: Categorical Cross Entropy
6. TESTING
Model is tested on unseen data to evaluate performance.
7. EVALUATION
Performance is measured using metrics:
• Accuracy
• Precision
• Recall
• F1-Score
Actual
Predicted
45 2 1
3 40 2
1 2 47
... ...
8. PREDICTION OUTPUT
System predicts the skin disease and provides confidence score.
Predicted Disease:
Psoriasis
Confidence Score:
92.4%
Goal: Provide accurate and automated diagnosis of 22 skin diseases using optimized ResNet-50 model.

Result

Model Performance

- Achieved **high classification accuracy**.
- Improved performance using optimized ResNet-50
- Reduced overfitting with augmentation & regularization

This graph shows that accuracy increases with epochs and stabilizes, indicating that the model is learning effectively without overfitting



Limitations

Yes, the current model has limitations, especially in recall. However, our focus was on implementing and optimizing the architecture. With more balanced data and further tuning, the performance can be significantly improved . And these are some following reasons or lessons we have learned during this project:-

- Imbalanced dataset (lower database of rare diseases)
- Low performance on rare diseases
- No class balancing technique (We used the dataset as it is to keep results realistic. Class balancing will be considered in future to improve recall)